

# Complex Analysis Qualifying Exam Problems

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# August 2018

## Part I

1. Find a conformal map  $f$  from the vertical strip

$$\Omega = \{z \in \mathbb{C} : 3 < \operatorname{Re}(z) < 11\}$$

onto the open unit disk  $\mathbb{D}$ .

*Solution.* The width of the strip is  $11 - 3 = 8$ . First map the strip to a horizontal strip by setting

$$w = \frac{\pi}{8}(z - 3).$$

Since  $3 < \operatorname{Re}(z) < 11$ , we have

$$0 < \operatorname{Re}(w) < \pi.$$

Now define

$$\omega = e^{iw}.$$

Because

$$\arg(\omega) = \operatorname{Re}(w),$$

the map  $\omega = e^{iw}$  sends the strip onto the upper half-plane

$$\{\omega \in \mathbb{C} : \operatorname{Im}(\omega) > 0\}.$$

Finally, the map

$$\phi(\omega) = \frac{\omega - i}{\omega + i}$$

maps the upper half-plane conformally onto the unit disk  $\mathbb{D}$ . Therefore a conformal map from  $\Omega$  onto  $\mathbb{D}$  is

$$f(z) = \frac{e^{i\pi(z-3)/8} - i}{e^{i\pi(z-3)/8} + i}$$

■

2. Suppose that  $F$  and  $G$  are holomorphic functions on the right half plane

$$H = \{z \in \mathbb{C} : \operatorname{Re}(z) > 0\}.$$

Suppose also that  $F$  and  $G$  have the same zeros with the same multiplicities at each zero. Prove that there is a holomorphic function  $h$  on  $H$  such that

$$F(z) = e^{h(z)}G(z)$$

for all  $z \in H$ .

*Solution.* If  $F \equiv G \equiv 0$ , then we may take  $h \equiv 0$ . Thus assume that  $F$  and  $G$  are not identically zero.

Define

$$q(z) = \frac{F(z)}{G(z)}$$

at points where  $G(z) \neq 0$ . We claim that  $q$  extends to a holomorphic function on all of  $H$ .

Let  $a \in H$  be a zero of  $G$ . Since  $F$  and  $G$  have the same zeros with the same multiplicities, there is some  $m \geq 1$  such that locally

$$F(z) = (z - a)^m \phi(z), \quad G(z) = (z - a)^m \psi(z),$$

where  $\phi$  and  $\psi$  are holomorphic near  $a$ , with

$$\phi(a) \neq 0, \quad \psi(a) \neq 0.$$

Hence near  $a$ ,

$$q(z) = \frac{F(z)}{G(z)} = \frac{\phi(z)}{\psi(z)}.$$

So the singularity of  $q$  at  $a$  is removable. Therefore  $q$  extends holomorphically to all of  $H$ .

Moreover, the extension of  $q$  has no zeros. Indeed, away from the zeros of  $G$ , this is clear since  $F$  and  $G$  have the same zeros. At a common zero  $a$ , we have

$$q(a) = \frac{\phi(a)}{\psi(a)} \neq 0.$$

Now  $H = \{z \in \mathbb{C} : \operatorname{Re}(z) > 0\}$  is convex, hence simply connected. Since  $q$  is holomorphic and never zero on  $H$ , by Corollary 6.17, Chapter IV, there exists a holomorphic function  $h$  on  $H$  such that

$$q(z) = e^{h(z)}.$$

Therefore, for all  $z \in H$ ,

$$F(z) = q(z)G(z) = e^{h(z)}G(z).$$

Thus there is a holomorphic function  $h$  on  $H$  satisfying

$$F(z) = e^{h(z)}G(z)$$

for all  $z \in H$ . ■

**3.** Suppose  $f$  is a holomorphic function on the open unit disc

$$D = \{z : |z| < 1\}.$$

Prove that there is a sequence  $(z_n)_{n=1}^{\infty}$  in  $D$  such that  $|z_n| \rightarrow 1$  and the set

$$\{f(z_n) : n \geq 1\}$$

is bounded.

*Solution (I).* If  $f \equiv 0$ , then take  $z_n = 1 - \frac{1}{n}$ . Then  $|z_n| \rightarrow 1$  and  $\{f(z_n) : n \geq 1\} = \{0\}$ , which is bounded.

Now assume  $f \not\equiv 0$ . Suppose, toward a contradiction, that no such sequence exists. Then for every  $M > 0$ , there is  $0 < \rho < 1$  such that

$$|f(z)| > M$$

whenever  $\rho < |z| < 1$ . Indeed, otherwise we could find  $z_n \in D$  with

$$|z_n| > 1 - \frac{1}{n} \quad \text{and} \quad |f(z_n)| \leq M,$$

which would give  $|z_n| \rightarrow 1$  and  $\{f(z_n)\}$  bounded. Thus

$$|f(z)| \rightarrow \infty \quad \text{as } |z| \rightarrow 1.$$

In particular, for some  $0 < R < 1$ ,

$$|f(z)| > 1$$

whenever  $R < |z| < 1$ . Hence all zeros of  $f$  lie in the compact set  $\{|z| \leq R\}$ . Since  $f \not\equiv 0$ , it has only finitely many zeros there. Let those zeros be  $\alpha_1, \dots, \alpha_N$ , with

multiplicities  $m_1, \dots, m_N$ , and define

$$P(z) = \prod_{j=1}^N (z - \alpha_j)^{m_j}.$$

If  $f$  has no zeros, take  $P \equiv 1$ . Then

$$F(z) = \frac{f(z)}{P(z)}$$

extends to a holomorphic function on  $D$ , and  $F$  has no zeros in  $D$ . Therefore

$$g(z) = \frac{1}{F(z)}$$

is holomorphic on  $D$ . Since  $P$  is bounded on  $D$ , and  $|f(z)| \rightarrow \infty$  as  $|z| \rightarrow 1$ , we get

$$|F(z)| \rightarrow \infty \quad \text{as } |z| \rightarrow 1.$$

Thus

$$g(z) \rightarrow 0 \quad \text{as } |z| \rightarrow 1.$$

Fix  $z_0 \in D$ . Given  $\varepsilon > 0$ , choose  $\rho < 1$  such that

$$|g(z)| < \varepsilon$$

whenever  $\rho < |z| < 1$ . Choose  $r$  such that

$$\max\{|z_0|, \rho\} < r < 1.$$

By the maximum modulus principle applied on  $\{|z| \leq r\}$ ,

$$|g(z_0)| \leq \max_{|z|=r} |g(z)| < \varepsilon.$$

Since  $\varepsilon > 0$  was arbitrary,  $g(z_0) = 0$ . Hence  $g \equiv 0$  on  $D$ , which is impossible because  $g = 1/F$ .

Therefore our supposition was false. Hence there exists a sequence  $(z_n) \subset D$  such that

$$|z_n| \rightarrow 1$$

and

$$\{f(z_n) : n \geq 1\}$$

is bounded. ■

*Solution (II).* Suppose, toward a contradiction, that no such sequence exists. Then

for every  $M > 0$ , there exists  $0 < \rho < 1$  such that

$$|f(z)| > M$$

whenever  $\rho < |z| < 1$ . Otherwise, for some  $M > 0$ , we could choose  $z_n \in D$  with

$$|z_n| > 1 - \frac{1}{n} \quad \text{and} \quad |f(z_n)| \leq M,$$

which would give the desired sequence. Thus

$$|f(z)| \rightarrow \infty \quad \text{as } |z| \rightarrow 1.$$

In particular, there exists  $0 < R < 1$  such that

$$|f(z)| > 1$$

whenever  $R < |z| < 1$ . Hence  $f$  has no zeros in the annulus

$$A = \{z : R < |z| < 1\}.$$

Therefore

$$g(z) = \frac{1}{f(z)}$$

is holomorphic on  $A$ . Also,

$$g(z) \rightarrow 0 \quad \text{as } |z| \rightarrow 1.$$

Now expand  $g$  in its Laurent series on the annulus  $A$ :

$$g(z) = \sum_{n=-\infty}^{\infty} c_n z^n.$$

For any  $r$  with  $R < r < 1$ ,

$$c_n = \frac{1}{2\pi i} \int_{|\omega|=r} \frac{g(\omega)}{\omega^{n+1}} d\omega.$$

Hence

$$|c_n| \leq \max_{|\omega|=r} |g(\omega)| r^{-n}.$$

Letting  $r \rightarrow 1^-$ , we have

$$\max_{|\omega|=r} |g(\omega)| \rightarrow 0,$$

and  $r^{-n} \rightarrow 1$ . Therefore

$$c_n = 0$$

for every  $n \in \mathbb{Z}$ . Hence

$$g \equiv 0$$

on  $A$ , which is impossible since  $g = 1/f$ . Therefore our supposition was false. Thus there exists a sequence  $(z_n) \subset D$  such that

$$|z_n| \rightarrow 1$$

and

$$\{f(z_n) : n \geq 1\}$$

is bounded. ■

**4.** Determine, with proof, all entire functions  $f$  such that for all  $\lambda \in \mathbb{C}$  with  $|\lambda| = 1$  and all  $z \in \mathbb{C}$  the equation

$$f(\lambda z) = \lambda f(z)$$

holds.

*Solution (I).* We claim that the only such functions are

$$f(z) = cz$$

for some constant  $c \in \mathbb{C}$ . Let  $f$  be entire. Then  $f$  has a power series expansion on all of  $\mathbb{C}$ :

$$f(z) = \sum_{n=0}^{\infty} a_n z^n.$$

For  $|\lambda| = 1$ , we have

$$f(\lambda z) = \sum_{n=0}^{\infty} a_n (\lambda z)^n = \sum_{n=0}^{\infty} a_n \lambda^n z^n.$$

By assumption,

$$f(\lambda z) = \lambda f(z) = \lambda \sum_{n=0}^{\infty} a_n z^n = \sum_{n=0}^{\infty} \lambda a_n z^n.$$

Therefore, for every  $|\lambda| = 1$ ,

$$\sum_{n=0}^{\infty} a_n \lambda^n z^n = \sum_{n=0}^{\infty} \lambda a_n z^n.$$

By uniqueness of power series coefficients,

$$a_n \lambda^n = \lambda a_n$$

for every  $n \geq 0$  and every  $|\lambda| = 1$ . Hence

$$a_n(\lambda^n - \lambda) = 0.$$

If  $n = 1$ , this condition gives no restriction on  $a_1$ . But if  $n \neq 1$ , we can choose  $|\lambda| = 1$  such that

$$\lambda^n \neq \lambda.$$

Thus  $a_n = 0$  for every  $n \neq 1$ . Therefore

$$f(z) = a_1 z.$$

Conversely, every function of the form  $f(z) = cz$  satisfies

$$f(\lambda z) = c\lambda z = \lambda cz = \lambda f(z).$$

Hence the entire functions satisfying the condition are exactly

$$f(z) = cz, \quad c \in \mathbb{C}.$$

■

*Solution (II).* First put  $z = 0$ . Then

$$f(0) = \lambda f(0)$$

for every  $|\lambda| = 1$ . Choosing  $\lambda \neq 1$ , we get

$$f(0) = 0.$$

Hence

$$g(z) = \frac{f(z)}{z}$$

is holomorphic on  $\mathbb{C} \setminus \{0\}$ , and the singularity at 0 is removable. Define

$$g(0) = f'(0).$$

Then  $g$  is entire. For  $z \neq 0$  and  $|\lambda| = 1$ ,

$$g(\lambda z) = \frac{f(\lambda z)}{\lambda z} = \frac{\lambda f(z)}{\lambda z} = \frac{f(z)}{z} = g(z).$$

Thus, by continuity,

$$g(\lambda z) = g(z)$$

for all  $z \in \mathbb{C}$  and all  $|\lambda| = 1$ . In particular, if  $|\zeta| = 1$ , then  $\zeta = \lambda \cdot 1$  for some  $|\lambda| = 1$ ,

so

$$g(\zeta) = g(1).$$

Thus  $g$  is constant on the unit circle. Write the Taylor expansion of  $g$  at 0:

$$g(z) = \sum_{n=0}^{\infty} a_n z^n.$$

By Cauchy's integral formula for Taylor coefficients,

$$a_n = \frac{1}{2\pi i} \int_{|\zeta|=1} \frac{g(\zeta)}{\zeta^{n+1}} d\zeta.$$

For  $n \geq 1$ , since  $g(\zeta) = g(1)$  on  $|\zeta| = 1$ , we get

$$a_n = \frac{g(1)}{2\pi i} \int_{|\zeta|=1} \frac{1}{\zeta^{n+1}} d\zeta = 0.$$

Therefore all Taylor coefficients of  $g$  except possibly  $a_0$  vanish. Hence  $g$  is constant on  $\mathbb{C}$ . Say

$$g(z) = c.$$

Then

$$f(z) = zg(z) = cz.$$

Conversely, if  $f(z) = cz$ , then

$$f(\lambda z) = c\lambda z = \lambda cz = \lambda f(z).$$

Therefore the desired functions are exactly

$$f(z) = cz, \quad c \in \mathbb{C}.$$

■

**5.** Let  $f$  and  $g$  be entire functions satisfying  $f(0) = g(0) \neq 0$  and

$$|f(z)| \leq |g(z)| \quad \forall z \in \mathbb{C}.$$

Show that  $f = g$ .

*Solution.* Since  $g(0) = f(0) \neq 0$ , there exists  $r > 0$  such that

$$g(z) \neq 0$$

for all  $z \in B(0, r)$ . Define

$$h(z) = \frac{f(z)}{g(z)}$$

on  $B(0, r)$ . Then  $h$  is holomorphic on  $B(0, r)$ . Also, by hypothesis,

$$|h(z)| = \frac{|f(z)|}{|g(z)|} \leq 1$$

for all  $z \in B(0, r)$ . Moreover,

$$h(0) = \frac{f(0)}{g(0)} = 1.$$

Thus

$$|h(0)| = 1.$$

So  $h$  attains its maximum modulus at the interior point 0. By the maximum modulus principle,  $h$  is constant on  $B(0, r)$ . Since  $h(0) = 1$ , we get

$$h \equiv 1$$

on  $B(0, r)$ . Therefore

$$f(z) = g(z)$$

for all  $z \in B(0, r)$ . Since  $f$  and  $g$  are entire and agree on a nonempty open set, the identity theorem implies that

$$f = g$$

on all of  $\mathbb{C}$ . ■

## Part II

**6.** Determine all entire functions that satisfy the functional equation

$$f(3z) = (1 - 3z)f(z).$$

Express the answer in terms of an infinite product.

*Solution.* Replacing  $z$  by  $z/3$  in the functional equation gives

$$f(z) = (1 - z)f\left(\frac{z}{3}\right).$$

Iterating this identity  $n$  times, we obtain

$$f(z) = \left( \prod_{k=0}^{n-1} \left( 1 - \frac{z}{3^k} \right) \right) f\left(\frac{z}{3^n}\right).$$

Since  $f$  is entire, it is continuous, and hence

$$\lim_{n \rightarrow \infty} f\left(\frac{z}{3^n}\right) = f(0).$$

Now define

$$P(z) = \prod_{k=0}^{\infty} \left( 1 - \frac{z}{3^k} \right).$$

This product converges uniformly on compact subsets of  $\mathbb{C}$ . Indeed, for  $|z| \leq R$ ,

$$\sum_{k=0}^{\infty} \left| \frac{z}{3^k} \right| \leq R \sum_{k=0}^{\infty} \frac{1}{3^k} = \frac{3R}{2} < \infty.$$

Therefore,  $P$  is entire. Letting  $n \rightarrow \infty$  in the iterated identity gives

$$f(z) = f(0) \prod_{k=0}^{\infty} \left( 1 - \frac{z}{3^k} \right).$$

Thus every entire solution is of the form

$$f(z) = C \prod_{k=0}^{\infty} \left( 1 - \frac{z}{3^k} \right), \quad C \in \mathbb{C}.$$

Conversely, suppose

$$f(z) = C \prod_{k=0}^{\infty} \left( 1 - \frac{z}{3^k} \right).$$

Then

$$\begin{aligned} f(3z) &= C \prod_{k=0}^{\infty} \left( 1 - \frac{3z}{3^k} \right) \\ &= C(1 - 3z) \prod_{k=1}^{\infty} \left( 1 - \frac{z}{3^{k-1}} \right) \\ &= (1 - 3z)f(z). \end{aligned}$$

Hence all entire solutions are

$$f(z) = C \prod_{k=0}^{\infty} \left( 1 - \frac{z}{3^k} \right), \quad C \in \mathbb{C}.$$

■

7. Let  $F(z)$  be holomorphic on a disc  $D(z_0, r)$  so that

$$F(z_0) = F'(z_0) = 0 \quad \text{and} \quad F''(z_0) \neq 0.$$

Show that there are two curves  $\Gamma_1$  and  $\Gamma_2$  passing through  $z_0$  that are orthogonal at  $z_0$  and so that the function  $F$  when restricted to  $\Gamma_1$  is real-valued and has a minimum at  $z_0$ , and when restricted to  $\Gamma_2$  it is also real-valued and has a maximum at  $z_0$ .

*Solution.* Since  $F(z_0) = F'(z_0) = 0$  and  $F''(z_0) \neq 0$ , the function  $F$  has a zero of order exactly two at  $z_0$ . Hence, in a sufficiently small neighborhood of  $z_0$ , we may write

$$F(z) = (z - z_0)^2 g(z),$$

where  $g$  is holomorphic and

$$g(z_0) = \frac{F''(z_0)}{2} \neq 0.$$

After shrinking the neighborhood if necessary,  $g$  has no zeros. Therefore,  $g$  admits a holomorphic square root  $h$ , so that

$$h(z)^2 = g(z).$$

Define

$$\phi(z) = (z - z_0)h(z).$$

Then

$$F(z) = \phi(z)^2.$$

Moreover,

$$\phi(z_0) = 0 \quad \text{and} \quad \phi'(z_0) = h(z_0) \neq 0.$$

Thus, by the holomorphic inverse function theorem,  $\phi$  is biholomorphic from some neighborhood  $U$  of  $z_0$  onto a neighborhood  $V$  of 0. Let

$$\psi = \phi^{-1}.$$

Choose  $\varepsilon > 0$  sufficiently small so that

$$(-\varepsilon, \varepsilon) \cup i(-\varepsilon, \varepsilon) \subseteq V,$$

and define the curves

$$\Gamma_1 = \{\psi(t) : -\varepsilon < t < \varepsilon\}$$

and

$$\Gamma_2 = \{\psi(it) : -\varepsilon < t < \varepsilon\}.$$

Both curves pass through  $z_0$ , since  $\psi(0) = z_0$ . If  $z = \psi(t) \in \Gamma_1$ , then

$$F(z) = \phi(z)^2 = t^2.$$

Thus  $F$  is real-valued on  $\Gamma_1$  and has a strict minimum at  $z_0$ . Similarly, if  $z = \psi(it) \in \Gamma_2$ , then

$$F(z) = \phi(z)^2 = (it)^2 = -t^2.$$

Thus  $F$  is real-valued on  $\Gamma_2$  and has a strict maximum at  $z_0$ . Finally, parametrizing the curves by

$$\gamma_1(t) = \psi(t) \quad \text{and} \quad \gamma_2(t) = \psi(it),$$

we obtain

$$\gamma_1'(0) = \psi'(0) \quad \text{and} \quad \gamma_2'(0) = i\psi'(0).$$

Multiplication by  $i$  rotates a vector through an angle of  $\pi/2$ . Therefore, the tangent vectors to  $\Gamma_1$  and  $\Gamma_2$  at  $z_0$  are orthogonal, and hence the two curves are orthogonal at  $z_0$ . ■

**8.** *Prove that if  $f$  is a non-constant bounded holomorphic function defined on the half plane*

$$\{z : \operatorname{Im} z < 1\},$$

*then there is some point on the real axis where the value of  $f$  is not real.*

*Solution.* Suppose, for contradiction, that

$$f(x) \in \mathbb{R} \quad \text{for every } x \in \mathbb{R}.$$

Define

$$g(z) = \overline{f(\bar{z})}.$$

Since  $f$  is holomorphic on

$$\{z \in \mathbb{C} : \operatorname{Im} z < 1\},$$

the function  $g$  is holomorphic on

$$\{z \in \mathbb{C} : \operatorname{Im} z > -1\}.$$

Both  $f$  and  $g$  are holomorphic on the strip

$$S = \{z \in \mathbb{C} : -1 < \operatorname{Im} z < 1\}.$$

For every  $x \in \mathbb{R}$ , we have

$$g(x) = \overline{f(\bar{x})} = f(x),$$

because  $f(x)$  is real. Since the real axis has accumulation points in  $S$ , the identity

theorem implies that

$$g(z) = f(z) \quad \text{for every } z \in S.$$

Therefore, the function

$$F(z) = \begin{cases} f(z), & \operatorname{Im} z < 1, \\ \overline{f(\bar{z})}, & \operatorname{Im} z > -1 \end{cases}$$

is well-defined and entire. Since  $f$  is bounded, there exists  $M > 0$  such that

$$|f(z)| \leq M \quad \text{whenever } \operatorname{Im} z < 1.$$

If  $\operatorname{Im} z > -1$ , then

$$\operatorname{Im}(\bar{z}) < 1,$$

so

$$|F(z)| = \left| \overline{f(\bar{z})} \right| = |f(\bar{z})| \leq M.$$

Thus  $F$  is a bounded entire function. By Liouville's theorem,  $F$  is constant. Hence  $f$  is constant, contradicting the hypothesis. Therefore, there exists some  $x \in \mathbb{R}$  such that

$$f(x) \notin \mathbb{R}.$$

■

**9.** Show that there exists a sequence of polynomials  $(P_n)$  so that

$$\lim_{n \rightarrow \infty} P_n(z) = \begin{cases} 1, & \text{if } \operatorname{Re}(z) > 0, \\ 0, & \text{if } \operatorname{Re}(z) = 0, \\ -1, & \text{if } \operatorname{Re}(z) < 0. \end{cases}$$

Also, explain why  $(P_n)$  can't be bounded on compact subsets of  $\mathbb{C}$ .

*Solution.* For each  $n \in \mathbb{N}$ , define

$$K_n^+ = \left\{ x + iy : \frac{1}{n} \leq x \leq n, |y| \leq n \right\},$$

$$K_n^- = \left\{ x + iy : -n \leq x \leq -\frac{1}{n}, |y| \leq n \right\},$$

and

$$K_n^0 = \{iy : |y| \leq n\}.$$

Let

$$K_n = K_n^+ \cup K_n^- \cup K_n^0.$$

The set  $K_n$  is compact, and its complement is connected. Define  $g_n : K_n \rightarrow \mathbb{C}$  by

$$g_n(z) = \begin{cases} 1, & z \in K_n^+, \\ -1, & z \in K_n^-, \\ 0, & z \in K_n^0. \end{cases}$$

The three sets  $K_n^+$ ,  $K_n^-$ , and  $K_n^0$  are pairwise disjoint, so  $g_n$  is continuous on  $K_n$ . Moreover,  $g_n$  is holomorphic on the interior of  $K_n$ . By Mergelyan's theorem, there exists a polynomial  $P_n$  such that

$$\sup_{z \in K_n} |P_n(z) - g_n(z)| < \frac{1}{n}.$$

Let  $z = x + iy \in \mathbb{C}$ . If  $x > 0$ , then for all sufficiently large  $n$ ,

$$\frac{1}{n} \leq x \leq n \quad \text{and} \quad |y| \leq n.$$

Thus  $z \in K_n^+$ , and hence

$$|P_n(z) - 1| < \frac{1}{n}.$$

Therefore,

$$\lim_{n \rightarrow \infty} P_n(z) = 1.$$

If  $x < 0$ , then for all sufficiently large  $n$ , we have  $z \in K_n^-$ , so

$$|P_n(z) + 1| < \frac{1}{n}.$$

Therefore,

$$\lim_{n \rightarrow \infty} P_n(z) = -1.$$

Finally, if  $x = 0$ , then  $z = iy$ , and for all sufficiently large  $n$ , we have  $z \in K_n^0$ . Hence

$$|P_n(z)| < \frac{1}{n},$$

and therefore

$$\lim_{n \rightarrow \infty} P_n(z) = 0.$$

Consequently,

$$\lim_{n \rightarrow \infty} P_n(z) = \begin{cases} 1, & \operatorname{Re}(z) > 0, \\ 0, & \operatorname{Re}(z) = 0, \\ -1, & \operatorname{Re}(z) < 0. \end{cases}$$

It remains to show that  $(P_n)$  cannot be bounded on compact subsets of  $\mathbb{C}$ . Suppose, for contradiction, that it were bounded on compact subsets. Thus, for every

compact set  $K \subseteq \mathbb{C}$ , there would exist a constant  $M_K$  such that

$$|P_n(z)| \leq M_K \quad \text{for every } z \in K \text{ and every } n.$$

The family  $\{P_n\}$  would then be locally bounded. By Montel's theorem, there would exist a subsequence  $(P_{n_k})$  converging uniformly on compact subsets of  $\mathbb{C}$  to an entire function  $P$ . On the other hand, every subsequence of  $(P_n)$  has the same pointwise limit

$$G(z) = \begin{cases} 1, & \operatorname{Re}(z) > 0, \\ 0, & \operatorname{Re}(z) = 0, \\ -1, & \operatorname{Re}(z) < 0. \end{cases}$$

Therefore,  $P = G$  on  $\mathbb{C}$ . This is impossible because  $P$  is entire and hence continuous, whereas  $G$  is discontinuous at every point of the imaginary axis. Thus,  $(P_n)$  cannot be bounded on compact subsets of  $\mathbb{C}$ . ■

**10.** Let  $\mathfrak{F}$  be the collection of holomorphic functions on the open unit disk

$$D = \{z : |z| < 1\}$$

satisfying

$$\iint_D |f| \, dx \, dy \leq 1.$$

Prove or disprove that  $\mathfrak{F}$  is a normal family.

# August 2020

## Part I

1. Let  $f$  be an entire function. Show that  $f$  is a constant function if there exist  $M > 0$  and  $R > 0$  such that

$$|f(z)| \leq M|z|^{1/2} \quad \text{whenever } |z| > R.$$

*Solution.* Expand  $f$  in its Taylor series about 0:

$$f(z) = \sum_{n=0}^{\infty} a_n z^n.$$

Let  $r > R$ . By Cauchy's estimate,

$$|a_n| \leq \frac{\max_{|z|=r} |f(z)|}{r^n}.$$

Since

$$|f(z)| \leq M|z|^{1/2}$$

whenever  $|z| > R$ , we have

$$\max_{|z|=r} |f(z)| \leq Mr^{1/2}.$$

Therefore,

$$|a_n| \leq Mr^{1/2-n}.$$

For every  $n \geq 1$ , we have  $1/2 - n < 0$ . Letting  $r \rightarrow \infty$ , we obtain

$$|a_n| = 0.$$

Thus  $a_n = 0$  for every  $n \geq 1$ . Hence

$$f(z) = a_0$$

for all  $z \in \mathbb{C}$ , and therefore  $f$  is constant. ■

2. Fix  $w = re^{i\theta} \neq 0$  and let  $\gamma$  be a piecewise smooth curve in  $\mathbb{C} \setminus \{0\}$  from 1 to  $w$ .

Show that there is an integer  $k$  such that

$$\int_{\gamma} \frac{1}{z} dz = \log r + i\theta + 2\pi ik.$$

*Solution.* Let  $\sigma$  be the path from 1 to  $r$  along the positive real axis, followed by the circular arc from  $r$  to  $w = re^{i\theta}$ . More precisely, the first part is given by

$$\sigma_1(t) = t, \quad 1 \leq t \leq r,$$

and the second part is given by

$$\sigma_2(t) = re^{it}, \quad 0 \leq t \leq \theta.$$

Then

$$\int_{\sigma_1} \frac{1}{z} dz = \int_1^r \frac{1}{t} dt = \log r,$$

and

$$\int_{\sigma_2} \frac{1}{z} dz = \int_0^\theta \frac{1}{re^{it}} ire^{it} dt = i\theta.$$

Therefore,

$$\int_{\sigma} \frac{1}{z} dz = \log r + i\theta.$$

Now let

$$\Gamma = \gamma * (-\sigma),$$

where  $-\sigma$  denotes the reverse of  $\sigma$ . Then  $\Gamma$  is a closed curve in  $\mathbb{C} \setminus \{0\}$ . Hence, by the winding number formula,

$$\int_{\Gamma} \frac{1}{z} dz = 2\pi ik$$

for some  $k \in \mathbb{Z}$ . Since

$$\int_{\Gamma} \frac{1}{z} dz = \int_{\gamma} \frac{1}{z} dz - \int_{\sigma} \frac{1}{z} dz,$$

we obtain

$$\int_{\gamma} \frac{1}{z} dz - (\log r + i\theta) = 2\pi ik.$$

Therefore,

$$\int_{\gamma} \frac{1}{z} dz = \log r + i\theta + 2\pi ik$$

for some  $k \in \mathbb{Z}$ . ■

**3.** (a) Let  $a > 1$ . Show that the equation

$$ze^{a-z} = 1$$

has precisely one root in the closed unit disk  $\{z : |z| = 1\}$ .

(b) Show that the root in part (a) is positive.

*Solution.* (a) Define

$$F(z) = ze^{a-z} - 1 \quad \text{and} \quad G(z) = ze^{a-z}.$$

On the unit circle  $|z| = 1$ , we have

$$|G(z)| = |z|e^{a-\operatorname{Re} z} = e^{a-\operatorname{Re} z}.$$

Since  $\operatorname{Re} z \leq 1$  and  $a > 1$ ,

$$|G(z)| \geq e^{a-1} > 1 = |F(z) - G(z)|.$$

Therefore, by Rouché's theorem,  $F$  and  $G$  have the same number of zeros in the open unit disk. Since  $e^{a-z} \neq 0$  for every  $z \in \mathbb{C}$ , the function

$$G(z) = ze^{a-z}$$

has exactly one zero, namely  $z = 0$ , and this zero is simple. Thus  $F$  has exactly one zero in the open unit disk. Moreover, the strict inequality above shows that  $F$  has no zeros on  $|z| = 1$ . Hence the equation

$$ze^{a-z} = 1$$

has precisely one root in the closed unit disk

$$\{z \in \mathbb{C} : |z| \leq 1\}.$$

(b) Consider the real-valued continuous function

$$h(x) = xe^{a-x} - 1, \quad 0 \leq x \leq 1.$$

We have

$$h(0) = -1 < 0$$

and

$$h(1) = e^{a-1} - 1 > 0,$$

since  $a > 1$ . By the intermediate value theorem, there exists  $x_0 \in (0, 1)$  such that

$$h(x_0) = 0.$$

Thus

$$x_0 e^{a-x_0} = 1.$$

Therefore  $x_0$  is a positive root lying in the unit disk. By the uniqueness established in part (a), it is the unique root in the closed unit disk. Hence that root is positive. ■

4. Using the maximum modulus principle, prove the Fundamental Theorem of Algebra.

*Solution.* Let

$$p(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_0, \quad a_n \neq 0,$$

be a nonconstant polynomial. We must show that  $p$  has a zero in  $\mathbb{C}$ . Suppose, for contradiction, that  $p(z) \neq 0$  for every  $z \in \mathbb{C}$ . Then

$$f(z) = \frac{1}{p(z)}$$

is an entire function.

Since  $p$  is a nonconstant polynomial,

$$|p(z)| \longrightarrow \infty \quad \text{as} \quad |z| \longrightarrow \infty.$$

Consequently,

$$|f(z)| = \frac{1}{|p(z)|} \longrightarrow 0 \quad \text{as} \quad |z| \longrightarrow \infty.$$

In particular, there exists  $R > 0$  such that

$$|f(z)| < |f(0)| \quad \text{whenever} \quad |z| > R.$$

Because  $f$  is continuous on the compact disk

$$\overline{D(0, R)} = \{z \in \mathbb{C} : |z| \leq R\},$$

the function  $|f|$  attains its maximum there. Thus there exists  $z_0 \in \overline{D(0, R)}$  such that

$$|f(z_0)| = \max_{|z| \leq R} |f(z)|.$$

Since  $0 \in \overline{D(0, R)}$ , we have

$$|f(z_0)| \geq |f(0)|.$$

On the other hand, for  $|z| > R$ ,

$$|f(z)| < |f(0)| \leq |f(z_0)|.$$

Therefore,  $|f|$  attains its maximum over all of  $\mathbb{C}$  at the point  $z_0$ . By the maximum modulus principle,  $f$  must be constant. Hence  $p = 1/f$  is also constant, contradicting the assumption that  $p$  is nonconstant. Therefore  $p$  must have at least one zero in  $\mathbb{C}$ . This proves the Fundamental Theorem of Algebra. ■

**5.** Let  $f$  be an entire function, taking real values on the real axis, and imaginary values on the imaginary axis. Show that  $f(z)$  is an odd function; that is,

$$f(-z) = -f(z)$$

for all  $z$ .

*Solution.* Since  $f$  is entire, it has a Taylor expansion about 0:

$$f(z) = \sum_{n=0}^{\infty} a_n z^n.$$

Because  $f$  takes real values on the real axis, we have

$$f(x) \in \mathbb{R} \quad \text{for all } x \in \mathbb{R}.$$

Therefore all derivatives of  $f$  at 0 are real, and hence

$$a_n = \frac{f^{(n)}(0)}{n!} \in \mathbb{R} \quad \text{for every } n \geq 0.$$

Now let  $z = iy$ , where  $y \in \mathbb{R}$ . Then

$$f(iy) = \sum_{n=0}^{\infty} a_n i^n y^n.$$

The real part of this expression is

$$\operatorname{Re} f(iy) = \sum_{k=0}^{\infty} a_{2k} (-1)^k y^{2k}.$$

Since  $f$  takes purely imaginary values on the imaginary axis,

$$\operatorname{Re} f(iy) = 0 \quad \text{for all } y \in \mathbb{R}.$$

Thus

$$\sum_{k=0}^{\infty} a_{2k} (-1)^k y^{2k} = 0 \quad \text{for all } y \in \mathbb{R}.$$

By uniqueness of power series coefficients,

$$a_{2k} = 0 \quad \text{for all } k \geq 0.$$

Hence the Taylor series of  $f$  contains only odd powers:

$$f(z) = \sum_{k=0}^{\infty} a_{2k+1} z^{2k+1}.$$

Therefore,

$$f(-z) = \sum_{k=0}^{\infty} a_{2k+1} (-z)^{2k+1} = - \sum_{k=0}^{\infty} a_{2k+1} z^{2k+1} = -f(z).$$

Thus  $f$  is an odd function. ■

## Part II

**6.** *Given entire functions  $f, g$  without common zeros, prove that there exist entire functions  $A, B$  such that*

$$Af + Bg = 1.$$

*Hint: Use the Mittag-Leffler Theorem.*

*Solution.* Assume first that  $f \not\equiv 0$ , and let  $\{a_n\}$  be the zeros of  $f$ , listed without repetition. Since  $f$  and  $g$  have no common zeros, we have

$$g(a_n) \neq 0$$

for every  $n$ .

For each  $n$ , let  $P_n$  be the principal part at  $a_n$  of the Laurent expansion of the meromorphic function

$$\frac{1}{fg}.$$

By the Mittag-Leffler theorem, there exists a meromorphic function  $h$  on  $\mathbb{C}$  whose only possible poles are the points  $a_n$ , and whose principal part at  $a_n$  is  $P_n$ . Define

$$B = fh$$

and

$$A = \frac{1}{f} - gh.$$

We claim that  $A$  and  $B$  are entire. Suppose that  $a_n$  is a zero of  $f$  of multiplicity

$m$ . Since  $g(a_n) \neq 0$ , the function  $1/(fg)$  has a pole of order  $m$  at  $a_n$ . Hence  $h$  has a pole of order at most  $m$  at  $a_n$ , so the singularity of  $fh$  at  $a_n$  is removable. Therefore,  $B = fh$  is entire. Moreover,  $h$  and  $1/(fg)$  have the same principal part at each  $a_n$ . Thus

$$\frac{1}{fg} - h$$

is holomorphic at every zero of  $f$ . Since

$$A = \frac{1}{f} - gh = g \left( \frac{1}{fg} - h \right),$$

the function  $A$  is holomorphic at every zero of  $f$ . Away from the zeros of  $f$ , it is clearly holomorphic. Hence  $A$  is entire. Finally,

$$Af + Bg = \left( \frac{1}{f} - gh \right) f + (fh)g = 1 - fgh + fgh = 1.$$

Therefore, there exist entire functions  $A$  and  $B$  such that

$$Af + Bg = 1.$$

If  $f \equiv 0$ , then  $g$  has no zeros, so  $1/g$  is entire. In that case, take

$$A = 0, \quad B = \frac{1}{g}.$$

■

**7.** If  $f$  is a nonconstant entire function, show that

$$(f^3 + f^2)(\mathbb{C}) = \mathbb{C}.$$

**8.** Suppose that  $G$  is a region in  $\mathbb{C}$ ,  $\{f_n\}$  is a sequence in  $H(G)$ ,  $f$  is a nonconstant function, and  $f_n \rightarrow f$  in  $H(G)$ . Let  $a \in G$  and  $b = f(a)$ . Show that there is a sequence  $\{a_n\}$  in  $G$  such that

$$(i) \quad a = \lim a_n,$$

$$(ii) \quad f_n(a_n) = b \text{ for sufficiently large } n.$$

**9.** Let

$$G = \{z : \operatorname{Re}(z) > 0\},$$

and  $f : G \rightarrow G$  be a one-one analytic function from  $G$  into  $G$ . Suppose  $f(a) = a$  for some positive real number  $a$ . Show that

$$|f'(a)| \leq 1.$$

**10.** Let  $0 < R_1 < R_2$  and  $\alpha \in \mathbb{R}$ , and let  $G$  be the annulus centered at the origin with radii  $R_1$  and  $R_2$ . Let

$$z^\alpha = \exp(\alpha \log z),$$

where  $\log z$  is a branch of the logarithm. Prove that  $z^\alpha$  is analytic in  $G$  if and only if  $\alpha$  is an integer.

# August 2022

## Part I

1. Let  $f$  be analytic on the open square

$$\Omega = \{z = x + iy : x, y \in (-1, 1)\}$$

and suppose that  $|f| \leq 1$  on  $\Omega$ . Show that

$$|f'(0)| < 1.$$

*Solution.* Let  $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$ . Since  $\mathbb{D} \subset \Omega$ , the restriction of  $f$  to  $\mathbb{D}$  is analytic and satisfies  $|f| \leq 1$ . If  $f$  is constant, then  $f'(0) = 0 < 1$ , so suppose that  $f$  is nonconstant. By the maximum modulus principle,

$$|f(z)| < 1 \quad (z \in \mathbb{D}),$$

and hence  $f : \mathbb{D} \rightarrow \mathbb{D}$ . Set  $a = f(0)$ . By the Schwarz–Pick lemma,

$$|f'(0)| \leq 1 - |f(0)|^2 = 1 - |a|^2.$$

Therefore, if  $a \neq 0$ , then

$$|f'(0)| < 1.$$

It remains to consider the case  $f(0) = 0$ . By Schwarz’s lemma,

$$|f'(0)| \leq 1.$$

Suppose, for contradiction, that  $|f'(0)| = 1$ . Then the equality case of Schwarz’s lemma implies that

$$f(z) = e^{i\theta}z$$

for all  $z \in \mathbb{D}$ , for some  $\theta \in \mathbb{R}$ . Since both  $f$  and  $z \mapsto e^{i\theta}z$  are analytic on the connected domain  $\Omega$  and agree on the open set  $\mathbb{D}$ , the identity theorem gives

$$f(z) = e^{i\theta}z$$

for all  $z \in \Omega$ . However,

$$z = \frac{3}{4} + \frac{3}{4}i \in \Omega,$$

while

$$|f(z)| = |z| = \frac{3\sqrt{2}}{4} > 1,$$

contradicting the assumption that  $|f| \leq 1$  on  $\Omega$ . Hence equality cannot occur, and therefore

$$|f'(0)| < 1.$$

■

**2.** Find a conformal map from the vertical strip

$$\{z \in \mathbb{C} : -2 < \operatorname{Re}(z) < 2\}$$

onto the open unit disk  $D(0, 1)$ . Give an explicit formula for your map.

**3.** Let  $a > 0$ . Compute the integral

$$\int_{-\infty}^{\infty} \frac{\sin(ax)}{x(x^2 + 1)} dx$$

by using residue theory.

**4.** Let  $f$  be entire, injective, and without fixed points in  $\mathbb{C}$ . Show that it must be of the form

$$f(z) = az + b, \quad z \in \mathbb{C},$$

for some complex scalars  $a, b$  with  $a \neq 0$ .

**5.** Let  $(f_n)$  be a sequence of holomorphic functions on a domain  $\Omega$  that converges to a function  $f$  locally uniformly on  $\Omega$ , and suppose that

$$f_n(\Omega) \subseteq \Omega$$

for each  $n = 1, 2, \dots$ . Show that either  $f(\Omega) \subseteq \Omega$ , or else  $f(\Omega)$  consists of a single point in  $\partial\Omega$ .

## Part II

6. Determine the number of solutions, counted with multiplicity, of the equation

$$z^6 + 7z^3 - 2z + 3 = 0$$

that satisfy  $1 < |z| < 2$ .

7. Construct an entire function  $f$  satisfying the following three conditions:

- (a)  $f$  has simple zeros at each  $\sqrt{n}$ ,  $n = 1, 2, \dots$ ;
- (b)  $f$  has double zeros at each  $\pm i\sqrt{k}$ ,  $k = 0, 1, 2, \dots$ ; and
- (c)  $f$  has no other zeros.

8. Prove or disprove: There exists a sequence of polynomials that converges point-wise to the Dirac function  $\delta$  on the complex plane, where

$$\delta(z) = \begin{cases} 1, & z = 0, \\ 0, & z \neq 0. \end{cases}$$

9. Let  $f$  be analytic on the open unit disk  $\mathbb{D}$  and continuous on its closure, and suppose that  $|f| = 1$  on the unit circle  $\partial\mathbb{D}$ . Find a formula for  $f$ .

Hint: Consider first the case when  $f$  is zero-free.

10. Show that, given any small  $\varepsilon > 0$ , for sufficiently large  $n$ , all the zeros of

$$f_n(z) = 1 + \frac{1}{z} + \frac{1}{2!z^2} + \cdots + \frac{1}{n!z^n}$$

must lie inside the disk  $D(0, \varepsilon)$ .

# August 2024

## Part I

1. Let  $G$  and  $\Omega$  be two regions in  $\mathbb{C}$ , and  $h : G \rightarrow \mathbb{C}$  be a one-one analytic function, and  $f : G \rightarrow \mathbb{C}$  be a continuous function with  $f(G) \subset \Omega$ , and  $g : \Omega \rightarrow \mathbb{C}$  be an analytic function with  $g'(w) \neq 0$  for any  $w \in \Omega$ . Show that  $f$  is analytic if

$$h(z) = g(f(z)), \quad \text{for all } z \in G.$$

*Solution (I).* Fix  $z_0 \in G$ , and set

$$w_0 = f(z_0) \in \Omega.$$

Since  $g'(w_0) \neq 0$ , the complex inverse function theorem implies that there exist neighborhoods  $U \subset \Omega$  of  $w_0$  and  $W$  of  $g(w_0)$  such that

$$g|_U : U \longrightarrow W$$

is one-to-one and has an analytic inverse

$$(g|_U)^{-1} : W \longrightarrow U.$$

Since  $f$  is continuous at  $z_0$ , there exists a neighborhood  $V \subset G$  of  $z_0$  such that

$$f(V) \subset U.$$

For every  $z \in V$ , we have

$$h(z) = g(f(z)) \in g(U) = W.$$

Therefore,

$$f(z) = (g|_U)^{-1}(h(z)), \quad z \in V.$$

Thus, on  $V$ ,  $f$  is the composition of two analytic functions:

$$f = (g|_U)^{-1} \circ h.$$

Hence  $f$  is analytic in a neighborhood of  $z_0$ . Since  $z_0$  was arbitrary,  $f$  is analytic on

$G$ . ■

*Solution (II).* First observe that  $f$  is one-to-one. Indeed, if

$$f(z_1) = f(z_2),$$

then

$$h(z_1) = g(f(z_1)) = g(f(z_2)) = h(z_2).$$

Since  $h$  is one-to-one, it follows that  $z_1 = z_2$ .

Fix  $a \in G$ . For  $z \neq a$ , the injectivity of  $f$  gives  $f(z) \neq f(a)$ , and hence

$$\frac{h(z) - h(a)}{z - a} = \frac{g(f(z)) - g(f(a))}{f(z) - f(a)} \frac{f(z) - f(a)}{z - a}.$$

Equivalently,

$$\frac{f(z) - f(a)}{z - a} = \frac{\frac{h(z) - h(a)}{z - a}}{\frac{g(f(z)) - g(f(a))}{f(z) - f(a)}}.$$

As  $z \rightarrow a$ , the continuity of  $f$  implies that

$$f(z) \rightarrow f(a).$$

Therefore,

$$\frac{g(f(z)) - g(f(a))}{f(z) - f(a)} \rightarrow g'(f(a)).$$

By hypothesis,

$$g'(f(a)) \neq 0.$$

Also, since  $h$  is analytic,

$$\frac{h(z) - h(a)}{z - a} \rightarrow h'(a).$$

Consequently,

$$\lim_{z \rightarrow a} \frac{f(z) - f(a)}{z - a} = \frac{h'(a)}{g'(f(a))}.$$

Thus  $f$  is complex differentiable at  $a$ , with

$$f'(a) = \frac{h'(a)}{g'(f(a))}.$$

Since  $a \in G$  was arbitrary,  $f$  is analytic on  $G$ . ■

**2.** Let  $D$  be the open unit disk. Show that no analytic function  $f : D \rightarrow \mathbb{C}$  has the property that

$$|f(z_n)| \rightarrow \infty, \quad \text{whenever } |z_n| \rightarrow 1.$$

*Solution.* Suppose, toward a contradiction, that such an analytic function  $f$  exists. First, the sequential hypothesis implies that

$$|f(z)| \longrightarrow \infty \quad \text{uniformly as } |z| \rightarrow 1.$$

More precisely, for every  $M > 0$ , there exists  $0 < r < 1$  such that

$$r < |z| < 1 \implies |f(z)| > M.$$

Indeed, otherwise there would exist  $M > 0$  and a sequence  $(z_n) \subset D$  such that

$$|z_n| \rightarrow 1 \quad \text{and} \quad |f(z_n)| \leq M,$$

contrary to the hypothesis. In particular, choose  $0 < r_0 < 1$  such that

$$|f(z)| > 1 \quad \text{whenever } r_0 < |z| < 1.$$

Thus all the zeros of  $f$  lie in the compact disk  $\{z : |z| \leq r_0\}$ . Since  $f \not\equiv 0$ , it has only finitely many zeros in  $D$ . Let these zeros, repeated according to multiplicity, be  $a_1, \dots, a_N$ , and set

$$P(z) = \prod_{j=1}^N (z - a_j).$$

If  $f$  has no zeros, take  $P \equiv 1$ . The function

$$F(z) = \frac{P(z)}{f(z)}$$

has removable singularities at the zeros of  $f$ , and hence extends to an analytic function on  $D$ . Moreover,  $F \not\equiv 0$ . Since  $P$  is bounded on  $D$ , while  $|f(z)| \rightarrow \infty$  as  $|z| \rightarrow 1$ , we have

$$|F(z)| = \frac{|P(z)|}{|f(z)|} \longrightarrow 0 \quad \text{as } |z| \rightarrow 1.$$

Therefore,

$$\max_{|z|=r} |F(z)| \longrightarrow 0 \quad \text{as } r \rightarrow 1^-.$$

Fix  $z \in D$ . For every  $r$  satisfying  $|z| < r < 1$ , the maximum modulus principle gives

$$|F(z)| \leq \max_{|\zeta|=r} |F(\zeta)|.$$

Letting  $r \rightarrow 1^-$ , we obtain  $F(z) = 0$ . Since  $z \in D$  was arbitrary,  $F \equiv 0$ , contradicting  $F \not\equiv 0$ . Hence no such analytic function  $f : D \rightarrow \mathbb{C}$  exists. ■

3. How many roots does the polynomial

$$z^9 + z^5 - 8z^3 + 2z + 1$$

have between the circles  $\{|z| = 1\}$  and  $\{|z| = 2\}$ ?

*Solution.* Let

$$P(z) = z^9 + z^5 - 8z^3 + 2z + 1.$$

We first count the zeros of  $P$  in the disk  $\{|z| < 2\}$ . On  $|z| = 2$ ,

$$|z^9| = 2^9 = 512,$$

whereas

$$|z^5 - 8z^3 + 2z + 1| \leq |z|^5 + 8|z|^3 + 2|z| + 1 = 32 + 64 + 4 + 1 = 101.$$

Thus

$$|z^5 - 8z^3 + 2z + 1| < |z^9| \quad \text{on } |z| = 2.$$

By Rouché's theorem,  $P(z)$  and  $z^9$  have the same number of zeros in  $\{|z| < 2\}$ . Hence  $P$  has 9 zeros in  $\{|z| < 2\}$ , counting multiplicities. Next, we count the zeros in  $\{|z| < 1\}$ . On  $|z| = 1$ ,

$$|-8z^3| = 8,$$

while

$$|z^9 + z^5 + 2z + 1| \leq |z|^9 + |z|^5 + 2|z| + 1 = 1 + 1 + 2 + 1 = 5.$$

Therefore,

$$|z^9 + z^5 + 2z + 1| < |-8z^3| \quad \text{on } |z| = 1.$$

Again, by Rouché's theorem,  $P(z)$  and  $-8z^3$  have the same number of zeros in  $\{|z| < 1\}$ . Thus  $P$  has 3 zeros in  $\{|z| < 1\}$ , counting multiplicities. Consequently, the number of zeros lying between the two circles is

$$9 - 3 = 6.$$

■

4. Prove Liouville's Theorem by considering the following integral

$$\int_{C_R} \frac{f(z)}{(z-a)(z-b)} dz,$$

for a bounded entire function  $f(z)$ , where  $C_R = Re^{it}$  with  $t \in [0, 2\pi]$ , and  $a \neq b$  with  $|a| < R$  and  $|b| < R$ .

*Solution.* Let  $f$  be a bounded entire function. Then there exists  $M > 0$  such that

$$|f(z)| \leq M \quad \text{for all } z \in \mathbb{C}.$$

Fix two distinct points  $a, b \in \mathbb{C}$ , and choose

$$R > \max\{|a|, |b|\}.$$

Consider

$$I_R = \int_{C_R} \frac{f(z)}{(z-a)(z-b)} dz,$$

where  $C_R = \{z : |z| = R\}$  is positively oriented. Using

$$\frac{1}{(z-a)(z-b)} = \frac{1}{a-b} \left( \frac{1}{z-a} - \frac{1}{z-b} \right),$$

Cauchy's integral formula gives

$$\begin{aligned} I_R &= \frac{1}{a-b} \left( \int_{C_R} \frac{f(z)}{z-a} dz - \int_{C_R} \frac{f(z)}{z-b} dz \right) \\ &= \frac{2\pi i}{a-b} (f(a) - f(b)). \end{aligned}$$

On the other hand, for  $z \in C_R$ , the reverse triangle inequality gives

$$|z-a| \geq R-|a|, \quad |z-b| \geq R-|b|.$$

Hence, by the  $ML$ -estimate,

$$\begin{aligned} |I_R| &\leq (2\pi R) \max_{|z|=R} \left| \frac{f(z)}{(z-a)(z-b)} \right| \\ &\leq \frac{2\pi RM}{(R-|a|)(R-|b|)}. \end{aligned}$$

As  $R \rightarrow \infty$ ,

$$\frac{2\pi RM}{(R-|a|)(R-|b|)} \rightarrow 0.$$

Therefore  $I_R \rightarrow 0$ . Since

$$I_R = \frac{2\pi i}{a-b} (f(a) - f(b))$$

is independent of  $R$ , it follows that

$$\frac{2\pi i}{a-b} (f(a) - f(b)) = 0.$$

Because  $a \neq b$ , we conclude that

$$f(a) = f(b).$$

The points  $a$  and  $b$  were arbitrary, so  $f$  is constant on  $\mathbb{C}$ . This proves Liouville's Theorem. ■

5. Let  $G$  be a region, and

$$G^* = \{z : \bar{z} \in G\}.$$

If  $f : G \rightarrow \mathbb{C}$  is analytic, prove that  $f^* : G^* \rightarrow \mathbb{C}$  defined by

$$f^*(z) = \overline{f(\bar{z})}$$

is also analytic.

*Solution.* The map  $z \mapsto \bar{z}$  is a homeomorphism of  $\mathbb{C}$ . Hence, since  $G$  is a region, its reflection

$$G^* = \{z \in \mathbb{C} : \bar{z} \in G\}$$

is also a region.

Fix  $z_0 \in G^*$ . Then  $\bar{z}_0 \in G$ . For  $z \in G^*$ ,  $z \neq z_0$ , we have

$$\begin{aligned} \frac{f^*(z) - f^*(z_0)}{z - z_0} &= \frac{\overline{f(\bar{z})} - \overline{f(\bar{z}_0)}}{z - z_0} \\ &= \frac{\overline{f(\bar{z}) - f(\bar{z}_0)}}{\bar{z} - \bar{z}_0}. \end{aligned}$$

As  $z \rightarrow z_0$ , we have

$$\bar{z} \rightarrow \bar{z}_0.$$

Since  $f$  is analytic at  $\bar{z}_0$ ,

$$\frac{f(\bar{z}) - f(\bar{z}_0)}{\bar{z} - \bar{z}_0} \rightarrow f'(\bar{z}_0).$$

Taking complex conjugates gives

$$\lim_{z \rightarrow z_0} \frac{f^*(z) - f^*(z_0)}{z - z_0} = \overline{f'(\bar{z}_0)}.$$

Thus  $f^*$  is complex differentiable at every  $z_0 \in G^*$ , and

$$(f^*)'(z) = \overline{f'(\bar{z})}.$$

Therefore  $f^*$  is analytic on  $G^*$ . ■

## Part II

6. Prove that a nonconstant harmonic function on a region is an open map.
7. Let  $G$  be a region and  $f, f_1, f_2, \dots$  be elements of  $H(G)$ . Show that  $f_n \rightarrow f$  in  $H(G)$  if and only if for each closed rectifiable curve  $\gamma$  in  $G$ ,  $f_n \rightarrow f$  uniformly on  $\gamma$ .
8. Let  $G$  be a region and suppose  $\mathcal{F}$  is normal in  $H(G)$  and  $\Omega$  is open in  $\mathbb{C}$  such that  $f(G) \subset \Omega$  for every  $f \in \mathcal{F}$ . Show that if  $g$  is analytic on  $\Omega$  and is bounded on bounded sets then

$$\{g \circ f : f \in \mathcal{F}\}$$

is normal.

*Solution.* Let

$$\mathcal{G} = \{g \circ f : f \in \mathcal{F}\}.$$

Since  $f(G) \subset \Omega$  for every  $f \in \mathcal{F}$ , each function  $g \circ f$  is analytic on  $G$ . We first show that  $\mathcal{F}$  is locally bounded. Let  $K$  be a compact subset of  $G$ . Suppose that  $\mathcal{F}$  were not uniformly bounded on  $K$ . Then there would exist functions  $f_n \in \mathcal{F}$  and points  $z_n \in K$  such that

$$|f_n(z_n)| > n.$$

Since  $\mathcal{F}$  is normal in  $H(G)$ , some subsequence  $\{f_{n_k}\}$  converges uniformly on compact subsets of  $G$  to a function  $f \in H(G)$ . In particular,  $f_{n_k} \rightarrow f$  uniformly on  $K$ . Hence, for all sufficiently large  $k$ ,

$$\sup_{z \in K} |f_{n_k}(z)| \leq \sup_{z \in K} |f(z)| + 1,$$

which contradicts

$$|f_{n_k}(z_{n_k})| > n_k.$$

Therefore, there exists  $M_K > 0$  such that

$$|f(z)| \leq M_K$$

for every  $f \in \mathcal{F}$  and every  $z \in K$ . Now consider the bounded subset

$$E_K = \{w \in \Omega : |w| \leq M_K\}$$

of  $\Omega$ . Since  $g$  is bounded on bounded subsets of  $\Omega$ , there exists  $C_K > 0$  such that

$$|g(w)| \leq C_K$$

for every  $w \in E_K$ . Consequently, for every  $f \in \mathcal{F}$  and every  $z \in K$ ,

$$|(g \circ f)(z)| = |g(f(z))| \leq C_K.$$

Thus  $\mathcal{G}$  is locally bounded on  $G$ . By Montel's theorem,  $\mathcal{G}$  is normal. ■

**9.** *Show there is an analytic function  $f$  on*

$$D = \{z : |z| < 1\}$$

*which is not analytic on any region  $G$  which properly contains  $D$ .*

*Solution.* Let

$$D = \{z \in \mathbb{C} : |z| < 1\},$$

and define

$$a_j = \left(1 - \frac{1}{j}\right) e^{ij}, \quad m_j = 1, \quad j \geq 1.$$

Since

$$|a_j| = 1 - \frac{1}{j} < 1,$$

we have  $a_j \in D$ . The points  $a_j$  are distinct because their moduli are distinct. Moreover,

$$|a_j| \longrightarrow 1,$$

so the sequence  $\{a_j\}$  has no limit point in  $D$ . By Theorem 5.15, there exists an analytic function  $f$  on  $D$  whose only zeros are the points  $a_j$ , each with multiplicity 1. We claim that  $f$  cannot be extended analytically to any region properly containing  $D$ . Suppose, toward a contradiction, that there exists a region  $G$  such that

$$D \subsetneq G$$

and an analytic function  $F$  on  $G$  satisfying

$$F|_D = f.$$

Since  $G$  is open and properly contains  $D$ , it contains a point  $w$  such that

$$|w| > 1.$$

Because  $G$  is a region, it is path connected. Hence there exists a continuous path

$$\gamma : [0, 1] \longrightarrow G$$

such that

$$\gamma(0) = 0 \quad \text{and} \quad \gamma(1) = w.$$

Since

$$|\gamma(0)| < 1 < |\gamma(1)|,$$

the intermediate value theorem implies that there exists  $t_0 \in (0, 1)$  such that

$$|\gamma(t_0)| = 1.$$

Set

$$\xi = \gamma(t_0).$$

Then

$$\xi \in G \cap \partial D.$$

Since  $\frac{1}{2\pi}$  is irrational, the sequence

$$\{e^{ij}\}_{j=1}^{\infty}$$

is dense in the unit circle. In fact, every tail of this sequence is dense. Therefore, there exists an increasing sequence of integers  $\{j_k\}$  such that

$$e^{ij_k} \longrightarrow \xi.$$

Since  $j_k \rightarrow \infty$ , we obtain

$$a_{j_k} = \left(1 - \frac{1}{j_k}\right) e^{ij_k} \longrightarrow \xi.$$

Furthermore,

$$F(a_{j_k}) = f(a_{j_k}) = 0$$

for every  $k$ . Thus the zeros of  $F$  have a limit point  $\xi \in G$ . By the identity theorem,

$$F \equiv 0$$

on  $G$ . Consequently,

$$f \equiv 0$$

on  $D$ , which is impossible because the only zeros of  $f$  are the points  $a_j$ . Therefore,  $f$  is analytic on  $D$  and cannot be extended analytically to any region properly containing  $D$ . ■

**10.** Find an entire function  $f$  such that for every integer  $n \geq 1$ ,  $f$  has  $n$  as a zero of order  $n$ . Find the most elementary solution possible.

# January 2025

## Part I

1. *Prove the Fundamental Theorem of Algebra by completing the following two steps for a nonconstant polynomial*

$$p(z) = a_0 + a_1z + \cdots + a_nz^n,$$

where  $a_n \neq 0$  and  $n \geq 1$ .

(a) *Show that there exists a nonzero polynomial  $q(z)$  such that*

$$\frac{1}{z} = \frac{p(0)}{zp(z)} + \frac{q(z)}{p(z)},$$

for  $z \neq 0$  and  $p(z) \neq 0$ .

(b) *By integrating over the positively oriented circle*

$$C_R = \{Re^{it} : 0 \leq t \leq 2\pi\},$$

show that  $p(z_0) = 0$  for some  $z_0 \in \mathbb{C}$ .

*Solution.* (a) Since

$$p(z) - p(0) = a_1z + a_2z^2 + \cdots + a_nz^n,$$

we may write

$$p(z) - p(0) = z(a_1 + a_2z + \cdots + a_nz^{n-1}).$$

Define

$$q(z) = a_1 + a_2z + \cdots + a_nz^{n-1}.$$

Then  $q$  is a polynomial. Moreover,  $q \neq 0$ , since the coefficient of  $z^{n-1}$  is  $a_n \neq 0$ .

For  $z \neq 0$  and  $p(z) \neq 0$ , we have

$$\begin{aligned} \frac{p(0)}{zp(z)} + \frac{q(z)}{p(z)} &= \frac{p(0) + zq(z)}{zp(z)} \\ &= \frac{p(0) + p(z) - p(0)}{zp(z)} \\ &= \frac{p(z)}{zp(z)} \\ &= \frac{1}{z} \end{aligned}$$

Therefore,

$$\frac{1}{z} = \frac{p(0)}{zp(z)} + \frac{q(z)}{p(z)}$$

- (b) Suppose, toward a contradiction, that  $p$  has no zeros in  $\mathbb{C}$ . Then  $p(0) \neq 0$ , and the function

$$\frac{q(z)}{p(z)}$$

is entire.

Integrating the identity from part (a) over  $C_R$ , we obtain

$$\int_{C_R} \frac{1}{z} dz = p(0) \int_{C_R} \frac{1}{zp(z)} dz + \int_{C_R} \frac{q(z)}{p(z)} dz$$

By Cauchy's theorem,

$$\int_{C_R} \frac{q(z)}{p(z)} dz = 0$$

Also,

$$\int_{C_R} \frac{1}{z} dz = 2\pi i$$

Hence

$$\int_{C_R} \frac{1}{zp(z)} dz = \frac{2\pi i}{p(0)} \tag{1}$$

We now estimate the integral as  $R \rightarrow \infty$ . For  $|z| = R$ ,

$$\begin{aligned} |p(z)| &= \left| a_n z^n + \sum_{k=0}^{n-1} a_k z^k \right| \\ &\geq |a_n| R^n - \sum_{k=0}^{n-1} |a_k| R^k \end{aligned}$$

Since the leading term dominates the lower-degree terms, there exists  $R_0 > 0$

such that, for every  $R \geq R_0$ ,

$$|p(z)| \geq \frac{|a_n|}{2} R^n \quad \text{whenever } |z| = R.$$

Therefore, by the  $ML$ -estimate,

$$\begin{aligned} \left| \int_{C_R} \frac{1}{zp(z)} dz \right| &\leq 2\pi R \max_{|z|=R} \frac{1}{|z| |p(z)|} \\ &\leq 2\pi R \frac{1}{R \left( \frac{|a_n|}{2} R^n \right)} \\ &= \frac{4\pi}{|a_n| R^n}. \end{aligned}$$

Since  $n \geq 1$ ,

$$\int_{C_R} \frac{1}{zp(z)} dz \longrightarrow 0 \quad \text{as } R \rightarrow \infty.$$

This contradicts (1), because

$$\frac{2\pi i}{p(0)} \neq 0.$$

Therefore,  $p$  must have a zero. Hence there exists  $z_0 \in \mathbb{C}$  such that

$$p(z_0) = 0.$$

This proves the Fundamental Theorem of Algebra. ■

**Remark 5.1.1.** *If Cauchy's integral formula is available, part (a) is not strictly necessary. Indeed, under the assumption that  $p$  has no zeros, the function*

$$g(z) = \frac{1}{p(z)}$$

*is entire, and Cauchy's integral formula gives directly*

$$\int_{C_R} \frac{1}{zp(z)} dz = \int_{C_R} \frac{g(z)}{z-0} dz = 2\pi i g(0) = \frac{2\pi i}{p(0)}.$$

*The decomposition in part (a) is therefore best viewed as a direct derivation of this special case using Cauchy's theorem. It isolates the singular term*

$$\frac{1}{zp(0)}$$

*from an analytic remainder whose integral over  $C_R$  is zero.*

2. Show that the equation

$$e^z = cz,$$

where  $c$  is a constant with  $|c| > e$ , has exactly one solution in the set

$$\{z \in \mathbb{C} : \operatorname{Re} z < 1\}.$$

*Solution.* Let

$$F(z) = e^z - cz.$$

Suppose that  $z$  is a solution of

$$e^z = cz$$

with  $\operatorname{Re} z < 1$ . Taking absolute values, we obtain

$$e^{\operatorname{Re} z} = |c||z|.$$

Therefore,

$$|z| = \frac{e^{\operatorname{Re} z}}{|c|} < \frac{e}{|c|} < 1,$$

since  $|c| > e$ . Thus, every solution satisfying  $\operatorname{Re} z < 1$  lies in the unit disk. On the circle  $|z| = 1$ , we have

$$|e^z| = e^{\operatorname{Re} z} \leq e < |c| = |cz|.$$

Hence, by Rouché's theorem, the functions

$$e^z - cz \quad \text{and} \quad -cz$$

have the same number of zeros in the unit disk, counted with multiplicity. The function  $-cz$  has exactly one zero in the unit disk, namely  $z = 0$ , with multiplicity one. Consequently,  $F(z) = e^z - cz$  also has exactly one zero in the unit disk.

Finally, if  $|z| < 1$ , then

$$\operatorname{Re} z \leq |z| < 1.$$

Thus, the unique zero of  $F$  in the unit disk belongs to the set

$$\{z \in \mathbb{C} : \operatorname{Re} z < 1\}.$$

Since every solution in this set must lie in the unit disk, the equation

$$e^z = cz$$

has exactly one solution in

$$\{z \in \mathbb{C} : \operatorname{Re} z < 1\}.$$

■

**3.** Show that if  $f : \mathbb{C} \rightarrow \mathbb{C}$  is a continuous function such that  $f$  is analytic off  $[-1, 1]$ , then  $f$  is an entire function.

*Solution.* We use Morera's theorem. Since  $f$  is continuous on  $\mathbb{C}$ , it suffices to show that

$$\int_{\partial T} f(z) dz = 0$$

for every triangle  $T \subset \mathbb{C}$ . First, let  $P$  be a polygon contained in the closed upper half-plane. For  $\varepsilon > 0$ , define

$$P_\varepsilon = P + i\varepsilon.$$

Then  $P_\varepsilon$  lies entirely in the open upper half-plane, where  $f$  is analytic. Hence, by Cauchy's theorem,

$$\int_{\partial P_\varepsilon} f(z) dz = 0.$$

Since  $f$  is continuous, letting  $\varepsilon \rightarrow 0^+$  gives

$$\int_{\partial P} f(z) dz = 0.$$

Similarly, by translating downward, the same conclusion holds for every polygon contained in the closed lower half-plane. Now let  $T$  be an arbitrary triangle, and define

$$P_+ = T \cap \{z \in \mathbb{C} : \operatorname{Im} z \geq 0\}, \quad P_- = T \cap \{z \in \mathbb{C} : \operatorname{Im} z \leq 0\}.$$

The sets  $P_+$  and  $P_-$  are polygons, possibly empty or degenerate. By the preceding argument,

$$\int_{\partial P_+} f(z) dz = 0, \quad \int_{\partial P_-} f(z) dz = 0.$$

When these two integrals are added, the contributions along the portion of the real axis separating  $P_+$  and  $P_-$  cancel, since that segment is traversed in opposite directions. Therefore,

$$\int_{\partial T} f(z) dz = \int_{\partial P_+} f(z) dz + \int_{\partial P_-} f(z) dz = 0.$$

Thus the integral of  $f$  around every triangle is zero. By Morera's theorem,  $f$  is analytic on all of  $\mathbb{C}$ . Therefore,  $f$  is entire. ■

**4.** Using Liouville's Theorem, prove the Fundamental Theorem of Algebra.

*Proof.* Suppose  $p(z) \neq 0$  for all  $z$ , and let

$$f(z) = [p(z)]^{-1}.$$

Then  $f$  is an entire function. If  $p$  is not constant, then

$$\lim_{z \rightarrow \infty} p(z) = \infty,$$

and hence

$$\lim_{z \rightarrow \infty} f(z) = 0.$$

In particular, there exists  $R > 0$  such that

$$|f(z)| < 1 \quad \text{if } |z| > R.$$

Since  $f$  is continuous on  $\overline{B}(0; R)$ , there exists a constant  $M$  such that

$$|f(z)| \leq M \quad \text{for } |z| \leq R.$$

Hence  $f$  is bounded and therefore must be constant by Liouville's theorem. It follows that  $p$  must be constant, contradicting our assumption. ■

**5.** Let  $G$  be a region and  $z_0 \in G$ . Suppose  $f : G \rightarrow \mathbb{C}$  is analytic and  $f'(z_0) \neq 0$ . Show that

$$\int_{\gamma} \frac{1}{f(z) - f(z_0)} dz = \frac{2\pi i}{f'(z_0)}$$

if  $\gamma = \gamma(t) = z_0 + re^{it}$  for  $t \in [0, 2\pi]$  and  $r > 0$  is small enough.

*Solution.* Since  $f$  is analytic at  $z_0$ , define

$$g(z) = \begin{cases} \frac{f(z) - f(z_0)}{z - z_0}, & z \neq z_0, \\ f'(z_0), & z = z_0. \end{cases}$$

The singularity at  $z_0$  is removable, so  $g$  is analytic in a neighborhood of  $z_0$ . Moreover,

$$g(z_0) = f'(z_0) \neq 0.$$

By continuity of  $g$ , there exists  $r > 0$ , sufficiently small, such that the closed disk

$$\overline{D(z_0, r)} \subset G$$

and

$$g(z) \neq 0 \quad \text{for all } z \in \overline{D(z_0, r)}.$$

Thus

$$h(z) = \frac{1}{g(z)}$$

is analytic on a neighborhood of  $\overline{D(z_0, r)}$ . For  $z \neq z_0$ , we have

$$f(z) - f(z_0) = (z - z_0)g(z),$$

and hence

$$\frac{1}{f(z) - f(z_0)} = \frac{h(z)}{z - z_0}.$$

Therefore, by Cauchy's integral formula,

$$\begin{aligned} \int_{\gamma} \frac{1}{f(z) - f(z_0)} dz &= \int_{\gamma} \frac{h(z)}{z - z_0} dz \\ &= 2\pi i h(z_0) \\ &= \frac{2\pi i}{g(z_0)} \\ &= \frac{2\pi i}{f'(z_0)}. \end{aligned}$$

Hence

$$\int_{\gamma} \frac{1}{f(z) - f(z_0)} dz = \frac{2\pi i}{f'(z_0)}$$

■

## Part II

**6.** Let  $G$  and  $\Omega$  be domains in the complex plane. If  $f : G \rightarrow \Omega$  is one-to-one and analytic and  $\varphi : \Omega \rightarrow \mathbb{R}$  is subharmonic, show that  $\varphi \circ f$  is subharmonic.

**7.** Let  $G$  be a domain in the complex plane and  $u : G \rightarrow \mathbb{R}$  be a nonconstant harmonic function and let

$$A = \{z \in G : u_x(z) = u_y(z) = 0\}.$$

Show that  $A$  does not have any limit points in  $G$ .

*Solution.* Define

$$f(z) = u_x(z) - i u_y(z).$$

Since  $u$  is harmonic on  $G$ , we have

$$u_{xx} + u_{yy} = 0.$$

Also, harmonic functions are smooth, so

$$u_{xy} = u_{yx}.$$

Thus, the real and imaginary parts of  $f$ , namely  $u_x$  and  $-u_y$ , satisfy the Cauchy–Riemann equations:

$$(u_x)_x = (-u_y)_y$$

and

$$(u_x)_y = -(-u_y)_x.$$

Therefore,  $f$  is holomorphic on  $G$ . Moreover,

$$A = \{z \in G : u_x(z) = u_y(z) = 0\} = \{z \in G : f(z) = 0\}.$$

Hence  $A$  is the zero set of the holomorphic function  $f$ . Suppose that  $A$  has a limit point in  $G$ . By the identity theorem,

$$f \equiv 0$$

on  $G$ . Consequently,

$$u_x \equiv 0 \quad \text{and} \quad u_y \equiv 0$$

on  $G$ . Since  $G$  is connected, this implies that  $u$  is constant on  $G$ , contradicting the assumption that  $u$  is nonconstant. Therefore,  $A$  has no limit points in  $G$ . ■

**8.** Find an entire function  $f$  such that

$$f(ne^{2k\pi i/n}) = 0$$

for all positive integers  $n$  and all integers  $0 \leq k \leq n-1$ . Give the most elementary example possible.

*Solution.* For each positive integer  $n$ , the points

$$ne^{2k\pi i/n}, \quad 0 \leq k \leq n-1,$$

are precisely the roots of the polynomial  $z^n - n^n$ . Thus, consider

$$f(z) = \prod_{n=1}^{\infty} \left(1 - \frac{z^n}{n^n}\right).$$

We first show that this product defines an entire function. Fix  $R > 0$ . For  $|z| \leq R$ , we have

$$\left| \frac{z^n}{n^n} \right| \leq \left( \frac{R}{n} \right)^n.$$

If  $n \geq 2R$ , then

$$\left( \frac{R}{n} \right)^n \leq 2^{-n}.$$

Consequently,

$$\sum_{n=1}^{\infty} \sup_{|z| \leq R} \left| \frac{z^n}{n^n} \right| \leq \sum_{n=1}^{\infty} \left( \frac{R}{n} \right)^n < \infty.$$

Therefore, the infinite product

$$\prod_{n=1}^{\infty} \left( 1 - \frac{z^n}{n^n} \right)$$

converges uniformly on every compact subset of  $\mathbb{C}$ . Since its partial products are polynomials,  $f$  is entire. Moreover,  $f(0) = 1$ , so  $f$  is not identically zero. Now let  $n$  be a positive integer and let  $0 \leq k \leq n-1$ . Then

$$(ne^{2k\pi i/n})^n = n^n e^{2k\pi i} = n^n.$$

Hence the  $n$ -th factor in the product vanishes:

$$1 - \frac{(ne^{2k\pi i/n})^n}{n^n} = 0.$$

It follows that

$$f(ne^{2k\pi i/n}) = 0.$$

Thus, an elementary example is

$$f(z) = \prod_{n=1}^{\infty} \left( 1 - \frac{z^n}{n^n} \right).$$

■

**9.** Let  $G$  and  $\Omega$  be two domains in the complex plane. Suppose  $\{f_n\}$  is a sequence in  $C(G, \Omega)$  which converges to  $f$  and  $\{z_n\}$  is a sequence in  $G$  which converges to a point  $z$  in  $G$ . Show

$$\lim_{n \rightarrow \infty} f_n(z_n) = f(z).$$

*Proof.* Since  $z_n \rightarrow z$  and  $z \in G$ , there exists  $r > 0$  such that

$$K = \overline{D(z, r)} \subset G.$$

Moreover,  $z_n \in K$  for all sufficiently large  $n$ . For such  $n$ , the triangle inequality gives

$$|f_n(z_n) - f(z)| \leq |f_n(z_n) - f(z_n)| + |f(z_n) - f(z)|.$$

Because  $f_n \rightarrow f$  uniformly on the compact set  $K$ ,

$$|f_n(z_n) - f(z_n)| \leq \sup_{w \in K} |f_n(w) - f(w)| \longrightarrow 0.$$

Also,  $f$  is continuous and  $z_n \rightarrow z$ , so

$$|f(z_n) - f(z)| \longrightarrow 0.$$

Therefore,

$$|f_n(z_n) - f(z)| \longrightarrow 0,$$

and hence

$$\lim_{n \rightarrow \infty} f_n(z_n) = f(z).$$

■

**10.** Show that if  $\mathcal{F} \subset H(G)$  is normal then

$$\mathcal{F}' = \{f' : f \in \mathcal{F}\}$$

is also normal.

*Solution.* Let  $\{g_n\}$  be an arbitrary sequence in  $\mathcal{F}'$ . For each  $n$ , choose  $f_n \in \mathcal{F}$  such that

$$g_n = f'_n.$$

Since  $\mathcal{F}$  is normal, there exist a subsequence  $\{f_{n_k}\}$  and a function  $f \in H(G)$  such that

$$f_{n_k} \longrightarrow f$$

uniformly on compact subsets of  $G$ . By Theorem VII.2.1, locally uniform convergence of analytic functions implies locally uniform convergence of all their derivatives. In particular,

$$f'_{n_k} \longrightarrow f'$$

uniformly on compact subsets of  $G$ . Since

$$g_{n_k} = f'_{n_k},$$

we obtain

$$g_{n_k} \longrightarrow f'$$

uniformly on compact subsets of  $G$ . Thus every sequence in  $\mathcal{F}'$  has a subsequence

converging uniformly on compact subsets of  $G$  to a function in  $H(G)$ . Therefore,  $\mathcal{F}'$  is normal. ■